
AB ≠ BA

Matrix multiplication does not commute.

If A is 3x5 and B is 5x4 then AB is 3x4 while BA is not defined.

If A is 3x5 and B is 5x3 then AB is 3x3 while BA is 5x5

If A is 3x3 and B is 3x3 then usually AB ≠ BA

Consider the entry at row 2, column 1, for AB

$$\text{Out}[*]= a_{21} b_{11} + a_{22} b_{21} + a_{23} b_{31}$$

Here it is for BA

$$\text{Out}[*]= a_{11} b_{21} + a_{21} b_{22} + a_{31} b_{23}$$

We can also search for simple examples.

$$\text{Out}[*]= \begin{array}{|c|c|c|} \hline \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} & \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} & \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \\ \hline \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \\ \hline \end{array}$$

There are also examples where AB = BA

$$\text{Out}[*]= \begin{array}{|c|c|c|} \hline \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \\ \hline \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} & \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \\ \hline \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} & \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} & \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \\ \hline \end{array}$$

the trivial case is where A = B so then syntactically AB = AA = BA, but also if A or B is either the identity matrix or the zero matrix.

There are other cases where AB = BA and the essential case is when they are diagonal matrices.

Diagonal Matrices

The elements off the diagonal are zero. The elements on the diagonal may or may not be zero.

MatrixForm[DiagonalMatrix[RandomInteger[{-2, 2}, 6]]]

Out[*]/MatrixForm=

$$\begin{pmatrix} 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Out[]//MatrixForm=

$$\begin{pmatrix} -2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 \end{pmatrix}$$

A diagonal matrix must be a square matrix, although in the general context it is defined by $a_{ij} = 0$ if $i \neq j$

The matrix product of diagonal matrices is simply the multiplication of the corresponding diagonal elements.

$$\text{Out[]}= \begin{pmatrix} -2 & 0 & 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} -2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Identity Matrix

A diagonal matrix with 1's on the diagonal.

The identity matrix is usually denoted by $\mathbb{I}$ when the dimension is understood.

Out[]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

For any matrix A (not necessarily square), the identity matrix multiplies like an identity element so that's what it is. For matrix algebra the identity matrix is the identity element.

$A \mathbb{I} = A = \mathbb{I} A$ (with the dimensions of $\mathbb{I}$ understood to be the rows/columns of A)

The entries of the identity matrix are the Kronecker delta δ_{ij}

So one can formally establish the above identities:

$$\text{Out[]}= \sum_{k=1}^n a_{ik} \delta_{kj} = a_{ij}$$

$$\text{Out[]}= \sum_{k=1}^n \delta_{ik} a_{kj} = a_{ij}$$

The identity satisfies

$$\det(I) = 1$$

$$\text{tr}(I) = n$$

Inverse Matrices

If A is a nonsingular square matrix then there is an inverse such that

$$A A^{-1} = I = A^{-1} A$$

The inverse of the identity is itself. That is not the only case where a matrix is itself. Here is an easy to see example.

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Usually the inverse is a different matrix.

Eigenvector Eigenvalue Equation

$$A v = \lambda v$$

A is a square matrix

v is called an eigenvector

λ is called an eigenvalue

The equation is trivial if $v = 0$ so we rule out the possibility of v being the zero vector.

An eigenvector is not the zero vector.

In this example, $\lambda = -3$

$$\begin{pmatrix} -2 & 1 \\ 2 & -1 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ -3 \end{pmatrix}$$

Eigenvector only defined up to a scalar multiplication

If v satisfies $A v = \lambda v$ for a fixed λ then so does

αv for any scalar α .

Recall that if v is an eigenvector it cannot be 0. So as long as $\alpha \neq 0$, then αv will also be an eigenvector.

$$\begin{pmatrix} -2 & 1 \\ 2 & -1 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ -3 \end{pmatrix}$$

In the above, if we double v then it is still an eigenvector. ($\alpha = 2$)

$$\text{Out}[*]= \begin{pmatrix} -2 & 1 \\ 2 & -1 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 2 \end{pmatrix} = \begin{pmatrix} 6 \\ -6 \end{pmatrix}$$

Or if we divide by -3 it is still an eigenvector ($\alpha = -1/3$)

$$\text{Out}[*]= \begin{pmatrix} -2 & 1 \\ 2 & -1 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{3} \\ -\frac{1}{3} \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Only special solutions

Generically a vector won't be an eigenvector for A because there is no λ that satisfies the equation.

For instance in this product A v

$$\text{Out}[*]= \begin{pmatrix} 0 & -1 \\ -2 & -2 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ 8 \end{pmatrix}$$

any vector of the form $\lambda \begin{pmatrix} -2 \\ -2 \end{pmatrix} = \begin{pmatrix} -2\lambda \\ -2\lambda \end{pmatrix}$ can never be equal to $\begin{pmatrix} 2 \\ 8 \end{pmatrix}$

On the other hand, generically any old number won't be an eigenvalue for A.

For instance, let v be an arbitrary nonzero vector and try $\lambda = 2$

$$\text{Out}[*]= \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x + y \\ -x \end{pmatrix}$$

Let's try to make this work for $\lambda=2$.

$\lambda \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$ can never be equal to $\begin{pmatrix} 2x + y \\ -x \end{pmatrix}$

unless both x and y are 0, i.e. v is the zero vector.

The system of equations is $(A - \lambda I) v = 0$

In order for there to be a nonzero solution to a homogeneous equations, with n equations and n variables, is for the matrix to be of lower rank.

Characteristic Polynomial

The system of equations is $(A - \lambda I) v = 0$ for the eigenvector eigenvalue equation.

To get a nonzero solution the rank of $A - \lambda I$ cannot be full.

That is one of the conditions for a matrix to be singular, and an equivalent condition is

$$\det(A - \lambda I) = 0$$

which is a polynomial of degree n in λ . Since there is no mention of the eigenvector v, this condition is only in terms of λ .

We call $\det(A - \lambda I)$ the characteristic polynomial $p(\lambda)$.

One solves $p(\lambda) = 0$ to find the eigenvalues.

Then one solves the homogeneous equation to get the eigenvector for that eigenvalue.

Examples:

Out[]=	$\begin{pmatrix} -1 & 0 & -2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$	$-x^3 - x^2$
	$\begin{pmatrix} 0 & 1 & -1 \\ 1 & 1 & -2 \\ 2 & 1 & -1 \end{pmatrix}$	$-x^3 - 2x - 2$
	$\begin{pmatrix} 1 & -1 & 1 \\ -1 & 0 & -2 \\ 0 & 1 & 2 \end{pmatrix}$	$-x^3 + 3x^2 - 3x - 1$

Out[]=	$\begin{pmatrix} 0 & 2 \\ -1 & -1 \end{pmatrix}$	$x^2 + x + 2$
	$\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$	$x^2 - 1$
	$\begin{pmatrix} 1 & 2 \\ 2 & 2 \end{pmatrix}$	$x^2 - 3x - 2$

Out[]=	$\begin{pmatrix} -2 & -2 & -2 & 0 \\ -2 & -2 & 1 & 1 \\ 1 & -2 & 0 & -2 \\ -1 & -1 & 1 & 0 \end{pmatrix}$	$x^4 + 4x^3 + 7x^2 + 30x + 12$
	$\begin{pmatrix} -1 & 2 & 1 & -2 \\ 2 & -2 & -2 & 2 \\ -1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 2 \end{pmatrix}$	$x^4 - 10x^2 + 12x$
	$\begin{pmatrix} 1 & -2 & 2 & 1 \\ -1 & -1 & -2 & -1 \\ -1 & 2 & 1 & 0 \\ 2 & 2 & 0 & 0 \end{pmatrix}$	$x^4 - x^3 + 3x^2 + 3x + 6$

Similar Matrices

A and B are square matrices of the same size.

A and B are similar if there is an invertible matrix C such that

Out[]= $C A C^{-1} = B$

The conjugation by C is equivalent to a change of basis.

Matrix Invariants

The determinant is not changed by conjugation, which can be seen using $\det(AB) = \det(A) \det(B)$

The determinant of the identity is 1

$\det(I) = 1$

Hence, $\det(C^{-1}) = 1/\det(C)$

so the determinant is the same after a change of basis.

Out[27]= $\det(C A C^{-1}) = \det(A)$

Consequently, the characteristic polynomial is invariant under change of basis, since

Out[29]= $C I C^{-1} = I$

so

Out[33]= $\det(C A C^{-1} - \lambda I) = \det(C A C^{-1} - C(\lambda I C^{-1})) = \det(A - \lambda I)$

The coefficients of the characteristic polynomial include the determinant and the trace.

We used the invariance of the determinant to show this so that isn't new but the trace is, as are the other coefficients.

Out[*]= $x^4 + a_{14}a_{23}a_{32}a_{41} - a_{13}a_{24}a_{32}a_{41} - a_{14}a_{22}a_{33}a_{41} + a_{12}a_{24}a_{33}a_{41} + a_{13}a_{22}a_{34}a_{41} - a_{12}a_{23}a_{34}a_{41} - a_{14}a_{23}a_{31}a_{42} + a_{13}a_{24}a_{31}a_{42} + a_{14}a_{21}a_{33}a_{42} - a_{11}a_{24}a_{33}a_{42} - a_{13}a_{21}a_{34}a_{42} + a_{11}a_{23}a_{34}a_{42} + a_{14}a_{22}a_{31}a_{43} - a_{12}a_{24}a_{31}a_{43} - a_{14}a_{21}a_{32}a_{43} + a_{11}a_{24}a_{32}a_{43} + a_{12}a_{21}a_{34}a_{43} - a_{11}a_{22}a_{34}a_{43} + x^3(-a_{11} - a_{22} - a_{33} - a_{44}) - a_{13}a_{22}a_{31}a_{44} + a_{12}a_{23}a_{31}a_{44} + a_{13}a_{21}a_{32}a_{44} - a_{11}a_{22}a_{32}a_{44} - a_{12}a_{21}a_{33}a_{44} + a_{11}a_{22}a_{33}a_{44} + x^2(-a_{12}a_{21} + a_{11}a_{22} - a_{13}a_{31} - a_{23}a_{32} + a_{11}a_{33} + a_{22}a_{33} - a_{14}a_{41} - a_{24}a_{42} - a_{34}a_{43} + a_{11}a_{44} + a_{22}a_{44} + a_{33}a_{44}) + x(a_{13}a_{22}a_{31} - a_{12}a_{23}a_{31} - a_{13}a_{21}a_{32} + a_{11}a_{23}a_{32} + a_{12}a_{21}a_{33} - a_{11}a_{22}a_{33} + a_{14}a_{22}a_{41} - a_{12}a_{24}a_{41} + a_{14}a_{33}a_{41} - a_{13}a_{34}a_{41} - a_{14}a_{21}a_{42} + a_{11}a_{24}a_{42} + a_{24}a_{33}a_{42} - a_{23}a_{34}a_{42} - a_{14}a_{31}a_{43} - a_{24}a_{32}a_{43} + a_{11}a_{34}a_{43} + a_{22}a_{34}a_{43} + a_{12}a_{21}a_{44} - a_{11}a_{22}a_{44} + a_{13}a_{31}a_{44} + a_{23}a_{32}a_{44} - a_{11}a_{33}a_{44} - a_{22}a_{33}a_{44})$

det	1	$a_{14}a_{23}a_{32}a_{41} - a_{13}a_{24}a_{32}a_{41} - a_{14}a_{22}a_{33}a_{41} + a_{12}a_{24}a_{33}a_{41} + a_{13}a_{22}a_{34}a_{41} - a_{12}a_{23}a_{34}a_{41} - a_{14}a_{23}a_{31}a_{42} + a_{13}a_{24}a_{31}a_{42} + a_{14}a_{21}a_{33}a_{42} - a_{11}a_{24}a_{33}a_{42} - a_{13}a_{21}a_{34}a_{42} + a_{11}a_{23}a_{34}a_{42} + a_{14}a_{22}a_{31}a_{43} - a_{12}a_{24}a_{31}a_{43} - a_{14}a_{21}a_{32}a_{43} + a_{11}a_{24}a_{32}a_{43} + a_{12}a_{21}a_{34}a_{43} - a_{11}a_{22}a_{34}a_{43} - a_{13}a_{22}a_{31}a_{44} + a_{12}a_{23}a_{31}a_{44} + a_{13}a_{21}a_{32}a_{44} - a_{11}a_{22}a_{32}a_{44} - a_{12}a_{21}a_{33}a_{44} + a_{11}a_{22}a_{33}a_{44}$
$-\sigma_3$	x	$a_{13}a_{22}a_{31} - a_{12}a_{23}a_{31} - a_{13}a_{21}a_{32} + a_{11}a_{23}a_{32} + a_{12}a_{21}a_{33} - a_{11}a_{22}a_{33} + a_{14}a_{22}a_{41} - a_{12}a_{24}a_{41} + a_{14}a_{33}a_{41} - a_{13}a_{34}a_{41} - a_{14}a_{21}a_{42} + a_{11}a_{24}a_{42} + a_{24}a_{33}a_{42} - a_{23}a_{34}a_{42} - a_{14}a_{31}a_{43} - a_{24}a_{32}a_{43} + a_{11}a_{34}a_{43} + a_{22}a_{34}a_{43} + a_{12}a_{21}a_{44} - a_{11}a_{22}a_{44} + a_{13}a_{31}a_{44} + a_{23}a_{32}a_{44} - a_{11}a_{33}a_{44} - a_{22}a_{33}a_{44}$
σ_2	x^2	$-a_{12}a_{21} + a_{11}a_{22} - a_{13}a_{31} - a_{23}a_{32} + a_{11}a_{33} + a_{22}a_{33} - a_{14}a_{41} - a_{24}a_{42} - a_{34}a_{43} + a_{11}a_{44} + a_{22}a_{44} + a_{33}a_{44}$
$-\text{tr}$	x^3	$-a_{11} - a_{22} - a_{33} - a_{44}$
	x^4	1

The invariants for a 4x4 matrix